XISOSIS

Deterministic Synergistic Non Archimedean Quantum Holonomy Automata

Revised mathematical foundations and quantum architecture for SU3 gauge theory on trees

DS NAQHA • Updated technical edition • 29 September 2026

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Abstract

We formulate a finite-tree SU(3) electric gauge model and separate its established mathematical properties from a proposed heterogeneous quantum computing architecture. Compact-group averaging gives exact gauge projectors, representation theory determines the electric spectrum, and finite-volume heat traces converge at positive temperature. Boundary conditions and external charges determine which representation states are physical. In particular, a fully constrained finite source-free tree has only the vacuum; a single fundamental flux link between static sources has energy $2g^2/3$ relative to the source-free reference, while suitable free-leaf boundary conditions permit a two-link excitation of energy $4g^2/3$.

The revised analysis does not infer an infinite-volume discrete spectrum, a universal physical mass gap, or an ultraviolet cutoff from compactness of the projective p-adic boundary. Quotient cycles allow magnetic interactions, and degenerate perturbation theory can produce nonzero first-order shifts even when diagonal matrix elements vanish. Scalar boundary scaling, register-size estimates and quantum subroutines are stated with their necessary hypotheses. A reproducible Python validation suite accompanies these results. The neutral-atom and superconducting architecture remains a research proposal, with no demonstrated transducer or hardware performance claimed.

To facilitate reproducibility, Python validation scripts accompany the mathematical results identified in Deterministic Synergistic Non Archimedean Quantum Holonomy Automata. The scripts implement the described algorithms and perform exact finite computations and numerical consistency checks supporting the analysis. These computational checks complement, but do not replace, the mathematical proofs and their stated assumptions. The code is available at:

https://github.com/teknoquants/Deterministic-Synergistic-Non-Archimedean-Quantum-Holonomy-Automata

I Introduction

Tree geometries provide a tractable setting for testing gauge constraints, electric spectra and quantum encodings. Their absence of independent cycles removes the usual magnetic plaquette dynamics. This simplification is useful for benchmarking, but it also limits what an open-tree model can establish about lattice gauge theories with loops or about four-dimensional Yang–Mills theory.

The DS NAQHA proposal combines SU(3) link variables on a Bruhat–Tits tree with a division of computational work between neutral-atom arrays and superconducting circuits. The mathematical model,

its computational realization and the device architecture require separate assessments. This edition retains that program while revising the ten numbered results and three subroutines to make their assumptions explicit.

Throughout, p is a prime, $g > 0$ is the gauge coupling, β is inverse temperature, and Δ denotes a scalar scaling dimension. We use (a,b) for SU(3) highest weights to avoid confusing representation labels with the prime. Natural logarithms define entropies. Energies follow the normalization of the electric Hamiltonian below; restoring physical lattice units requires a specified discretization.

II Geometry and invariant measure

Tree and boundary

Let T_p be the infinite regular tree of degree p+1. Its boundary is identified with the projective line $P^1(Q_p)$. The projective boundary is compact in its projective topology, whereas Q_p itself is not compact and the infinite tree is unbounded in graph distance. The inequality $|x|_p \le 1$ characterizes Z_p, not all of Q_p. For example, $|1/p|_p = p$.

A finite ball $T_{p,D}$ contains the root and all vertices at graph distance at most D. Its vertex and edge counts are

$$N_D \;=\; 1 \;+\; (p+1)\frac{p^D \;-\; 1}{p \;-\; 1}, \quad M_D \;=\; N_D \;-\; 1.$$

A rooted p-ary subtree has a different count, $(p^{D+1}-1)/(p-1)$, although both counts scale exponentially with depth. These two truncations should not be interchanged when computing exact resource estimates.

Link variables and gauge action

Choose one orientation for each undirected edge and denote this independent set by E^+. Assign $U_e \in SU(3)$ to $e \in E^+$ and set the variable on the reversed edge equal to $U_e\dagger$. Reverse orientations do not introduce additional independent degrees of freedom. Vertex transformations act as

$$U_{(u,v)} \;\mapsto\; g_u U_{(u,v)} {g_v}^{\dagger}.$$

The normalized product Haar measure on a finite graph is defined by the independent edges. Compactness of SU(3), rather than compactness of the spatial boundary, supplies the normalized link measure.

$$\mathcal{D}U \;=\; \prod_{e \,\in\, E^+} dU_e, \qquad \int_{SU(3)} dU \;=\; 1.$$

The fundamental matrix-entry identities are

$$\int_{SU(3)} U_{ij} {U^{\dagger}}_{kl}\, dU \;=\; \frac{1}{3}\delta_{il}\delta_{jk},$$

$$\int_{SU(3)} U_{i_1 j_1} U_{i_2 j_2} U_{i_3 j_3}\, dU \;=\; \frac{1}{6}\varepsilon_{i_1 i_2 i_3}\varepsilon_{j_1 j_2 j_3}.$$

The countable product measure is determined by its finite cylinder measures. It does not imply bounded electric energy: the Casimir eigenvalues grow without bound even on one compact-group link.

III Electric dynamics and physical sectors

For a finite graph, the unconstrained link Hilbert space is the tensor product of $L^2(SU(3))$ over E^+. In the electric lattice formulation [4], with generators normalized by $\mathrm{Tr}(T^a T^b)=\delta^{ab}/2$, define the local gauge generator and electric Hamiltonian by

$$G_v{}^a = \sum_{e\ out\ of\ v} L_e{}^a - \sum_{e\ into\ v} R_e{}^a,$$

$$H_E = \frac{g^2}{2} \sum_{e \in E^+} \sum_{a=1}^{8} (L_e{}^a)^2.$$

Here $L_e{}^a$ and $R_e{}^a$ denote the outgoing and incoming Lie generators in the chosen convention. Without external charges, physical states lie in the joint kernel of the constrained vertex generators. With static charges, the corresponding charge generators must be included in Gauss law. On an open tree the magnetic plaquette sum is empty. This says that H_B is absent, not that every closed-walk holonomy has zero trace: a backtracking walk has identity holonomy.

Peter–Weyl decomposition labels each link by a highest weight (a,b), with dimension and quadratic Casimir

$$d(a,b) = \frac{(a+1)(b+1)(a+b+2)}{2}, \quad C_2(a,b) = \frac{a^2}{3} + \frac{b^2 + ab + 3a + 3b}{3}.$$

The link energy is $g^2C_2(a,b)/2$. The trivial representation has zero energy, the fundamental and conjugate fundamental have $C_2=4/3$, the sextet has $C_2=10/3$, and the adjoint has $C_2=3$. Gauss constraints select admissible intertwiners between incident representations.

Sector or boundary condition	Lowest energy statement
Finite source-free tree with Gauss law at every vertex	Only the trivial state survives. There is no nontrivial excitation in this physical space.
Adjacent static fundamental and conjugate sources	Minimal connecting flux costs $2g^2/3$ relative to the source-free reference vacuum.
Finite regular-tree ball with unconstrained leaves and constrained interior	For $D \geq 1$, a shortest nonzero boundary flux path joins sibling leaves and costs $4g^2/3$.

These statements refer to different Hilbert spaces or boundary conditions. A charged-sector ground energy is not a first excited level above the source-free vacuum within the same sector. No continuum or infinite-volume physical mass-gap theorem follows from these finite-tree calculations.

IV Proposed heterogeneous architecture

The architecture assigns local vertex operations to neutral-atom arrays and branch communication or selected link operations to superconducting circuits. Rubidium-87 optical-tweezer arrays, hyperfine qubit states and Rydberg interactions remain candidate ingredients for the vertex subsystem. Transmon qubits and coplanar microwave resonators remain candidate ingredients for the branch subsystem. These are proposed roles, not experimentally established implementations of the present gauge model.

Two atomic hyperfine states or two transmon levels form a qubit; they do not alone encode a fundamental SU(3) qutrit or $L^2(SU(3))$. A finite implementation must specify an irrep truncation, a multi-qubit code, gauge-generator circuits and a method to suppress or detect leakage. Rydberg blockade alone is not a proof that all eight SU(3) Gauss constraints are enforced.

A realistic comparison of platforms requires gate counts, connectivity, transport overhead, coherence budgets and measured errors for the selected encoding. A hardware allocation based on tree adjacency can organize that study, but it does not establish a computational advantage or fault tolerance.

V Proposed quantum state transfer

A coherent interface would need to map the encoded atomic state to an optical mode, couple it to a pumped conversion device, transfer the quantum amplitude to a microwave mode and absorb it in a superconducting register. The photon frequency at 780 nm is approximately 384.35 THz. A 5.2 GHz microwave photon has roughly 74 thousand times less energy, so coherent frequency conversion requires an energy source or sink and a specified pump-assisted interaction.

A direct narrative in which one optical photon simply becomes one mechanical phonon and then one microwave photon is not a sufficient physical model. A quantitative proposal must specify the participating optical, mechanical and microwave modes, pump frequencies, detunings, coupling rates, loss channels, added noise and state-transfer fidelity. No such device model or measured conversion data is asserted here.

The research objective is coherent transfer between encoded registers, including phase preservation, heralding or error detection where appropriate, and compatibility with the computation's timing budget. The terms synergistic and deterministic fault-tolerant quantum computing describe intended architectural goals; they do not certify fault tolerance or eliminate measurement and synthesis errors.

VI Revised mathematical results

The following ten results retain the numbering of the original addendum. Their titles and statements have been revised where a numerical check or direct mathematical argument exposed a missing hypothesis or an invalid inference. Proofs below carry the mathematical conclusions; the accompanying scripts provide independent finite checks and counterexamples.

Result 1 Orthogonal gauge projection

Let U_v be a strongly continuous unitary representation of compact SU(3) on the local Hilbert space. The normalized Haar average P_v is the orthogonal projector onto its invariant subspace.

$$P_v = \int_{SU(3)} U_v(h)\, dh, \quad {P_v}^2 = P_v = {P_v}^\dagger.$$

Proof. The representation identity $U(g)U(h)=U(gh)$, Haar invariance and normalization give $P_v^2=P_v$. Invariance under inversion gives $P_v\dagger=P_v$. Left multiplication shows that every vector in the range is invariant, while averaging fixes every invariant vector. For this connected Lie group, invariant vectors are precisely those annihilated by all infinitesimal gauge generators, with the usual generator-domain interpretation. The argument also applies after adding charge representations to the local gauge action.

The scripts compare Haar averaging with the Gauss nullspace in fundamental, mesonic and baryonic sectors. They check idempotence, Hermiticity and annihilation by all eight generators.

Result 2 Finite volume spectrum and conditional flux energies

On finitely many independent links, H_E has a nonnegative discrete spectrum with finite multiplicities and compact resolvent. The same statement holds on a closed gauge-invariant reducing subspace. Its allowed energies are sums of $g^2C_2(a_e,b_e)/2$ over links, subject to the sector's intertwiner conditions.

Proof. Peter–Weyl decomposes each link into finite-dimensional representation blocks of multiplicity $d(a,b)^2$. The Casimir grows quadratically with the weights, so only finitely many label tuples have energy below any fixed bound on a finite graph. Each such tuple has finite multiplicity. The all-trivial tuple gives the unique unconstrained vacuum. Gauge restriction retains a subset of these eigenspaces.

For a finite source-free tree constrained at every vertex, a leaf has only one incident link and therefore forces that link to be trivial. Iterating this argument leaves only the vacuum. With an adjacent static source pair, the connecting fundamental flux instead costs $2g^2/3$. With free leaves on a regular-tree ball of depth at least one, a single excited link fails an interior Gauss constraint, while a two-link path between sibling leaves admits a singlet and costs $4g^2/3$.

These are finite-volume statements. In an unconstrained infinite-link reference-vacuum space, translating one excitation among infinitely many links gives infinitely many orthogonal states at the same energy. That example prevents an inference of compact resolvent from finite-volume spectra. An infinite physical sector and its operator domain require their own construction.

Result 3 Vanishing bare open line expectation

Let γ be a nonempty simple path, and let D^r be a nontrivial irreducible SU(3) representation. In the factorized constant vacuum, the expectation of the traced path holonomy vanishes.

$$\langle W_r(\gamma)\rangle \;=\; Tr \prod_{e\,\in\,\gamma}\Big[\int_{SU(3)} D^r(U)\,dU\Big] \;=\; 0.$$

Proof. Every nontrivial irreducible representation has zero Haar average because its invariant subspace is zero. Independent link integration factorizes the matrix product before taking its trace. At least one factor is present, so the result is zero. The same argument handles reversed orientations through U^{-1}. The open trace is generally gauge variant under independent endpoint transformations; its vanishing expectation does not make the operator identically zero.

Result 4 Cycles permit a magnetic multiplication operator

For a finite connected graph or finite connected quotient core, the cycle rank is $|E|-|V|+1$. A chosen nontrivial cycle can support a nonzero magnetic trace multiplication operator. On a tree the independent-cycle set is empty.

Proof. The oriented incidence matrix of a connected finite graph has rank $|V|-1$, so the kernel dimension is $|E|-|V|+1$. On a simple cycle with holonomy U_c, the character $\chi_3(U_c)$ is a nonzero function. Its multiplication operator sends the constant function to χ_3. Character orthogonality yields the nonzero transition matrix element below, even though its bare vacuum expectation is zero.

$$\int \chi_3{}^*(U)\,\chi_3(U)\,dU \;=\; 1, \qquad \int \chi_3(U)\,dU \;=\; 0.$$

A Schottky quotient can motivate such cycles, but the arithmetic group action, the quotient core and the chosen magnetic interactions must be specified. The expression $|E|-|V|+1$ cannot be used as arithmetic on

two infinite cardinalities. The magnetic term is an added modeling choice on selected cycles, not a nonzero bare Haar integral forced by topology.

Result 5 Scalar Green kernel and boundary scaling

Set $L=(p+1)I-A$ on the regular infinite tree, where A is adjacency. For $\Delta>1/2$ define the mass parameter and radial kernel by

$$m^2 = p^{\Delta} + p^{1-\Delta} - (p+1), \quad G(d) = \frac{p^{-\Delta d}}{p^{\Delta} - p^{-\Delta}}.$$

Then G is the decaying ℓ^2 resolvent kernel satisfying $(L+m^2)G=\delta$. The mass parameter may be negative; for $\Delta>1/2$ it lies above the lower stability threshold needed for this resolvent.

Proof. Away from the source, one neighbor reduces the distance and p neighbors increase it. Substitution gives $[(p+1)+m^2]G(d)-G(d-1)-pG(d+1)=0$. At the source, $[(p+1)+m^2]G(0)-(p+1)G(1)=1$ fixes the normalization. A shell has order p^d vertices, so square summability requires $p^{1-2\Delta}<1$. Equivalently, the sum of p raised to Δ and p raised to $1-\Delta$ exceeds $2\sqrt{p}$, placing the adjacency resolvent outside its spectral interval. The local recurrence still holds for $0<\Delta\leq 1/2$, but does not by itself give the same ℓ^2 resolvent interpretation.

In a Z_p boundary chart, let distinct x and y separate at depth v and regulate both endpoints at depth $N>v$. Then $|x-y|_p=p^{-v}$ and the geodesic length is $2(N-v)$. Removing the depth factor gives

$$p^{2\Delta N} G(2(N-v)) = \frac{|x-y|_p^{-2\Delta}}{p^{\Delta} - p^{-\Delta}}.$$

This is a scalar two-point scaling statement [1]. If the opposite Laplacian convention $\Box=-L$ is used, the displayed positive kernel obeys $(\Box-m^2)G=-\delta$. A full SU(3) operator duality does not follow from this scalar calculation.

Result 6 First order shifts require degenerate perturbation theory

On an open tree $H_B=0$, so every magnetic perturbative correction vanishes. On a graph with selected cycles, take

$$H_B = -\frac{1}{2g^2}\sum_c (\chi_3(U_c) + \chi_3^*(U_c)).$$

Its prefactor relative to the electric prefactor scales as g^{-4}. Some diagonal first-order expectations vanish by character selection rules. This does not imply zero first-order shifts in a degenerate eigenspace.

Counterexample. On a single cycle of length ℓ, the normalized gauge-invariant states χ_3 and $\chi_{\bar{3}}$ have equal electric energy $\ell g^2(4/3)/2$. In that subspace, the diagonal entries of H_B vanish, but the off-diagonal entries equal $-1/(2g^2)$. The fusion $3\otimes 3=\bar{3}\oplus 6$ supplies the nonzero coefficient. Consequently the symmetric and antisymmetric combinations have first-order shifts $-1/(2g^2)$ and $+1/(2g^2)$, respectively. The vacuum has zero first-order diagonal shift. Therefore the proposed general zero first-order gap shift on quotient cycles is false.

Any nontrivial strong-coupling analysis must diagonalize the perturbation within each degenerate electric eigenspace and specify the cycles, boundary conditions and charge sector. The tree statement remains exact but contains no magnetic dynamics.

Result 7 Product measure invariance under automorphisms

A tree automorphism preserves the product Haar measure when its action on the independent edge variables includes inversion whenever an orientation reverses.

Proof. On any finite cylinder function, the pullback relabels the independent integration variables and may replace some U_e by U_e^{-1}. Product structure, permutation invariance and Haar invariance under inversion preserve the integral. Equality on finite cylinders establishes equality of the countable product measures. The automorphism need not preserve the particular finite subset originally chosen for the cylinder.

Result 8 Register sizes under explicit truncation

A binary vertex-index register for N_D vertices needs $\lceil \log_2 N_D \rceil$ qubits, which is $O(D \log_2 p)$ for the finite regular-tree ball. A one-hot register instead needs N_D qubits. These are storage statements, not bounds on oracle or total gate complexity.

For a finite irrep set Λ, the truncated regular link space has dimension

$$d_\Lambda = \sum_{(a,b) \in \Lambda} d(a,b)^2, \quad q_{gauge} = M_D \lceil log_2 d_\Lambda \rceil.$$

Proof. Binary indexing requires enough basis strings to distinguish N_D vertices. Peter–Weyl supplies $d(a,b)^2$ states per link irrep, and an independent per-link qubit register needs $\lceil \log_2 d_\Lambda \rceil$ qubits. Multiplying by the number of links gives the displayed storage count. This is an uncompressed encoding; constraint-adapted encodings can change it. Even at fixed link truncation, independent-link storage grows exponentially in D. Efficient block-encoding queries require additional circuit and data-access assumptions.

Result 9 Finite volume heat trace convergence

For $g>0$, $\beta>0$ and finitely many links, the unprojected electric heat trace is finite and factorizes as

$$Z_{full} = K(t)^{M_D}, \quad t = \frac{\beta g^2}{2}, \quad K(t) = \sum_{a,b \geq 0} d(a,b)^2 e^{-tC_2(a,b)}.$$

Proof. Let $n=a+b$. Since $ab \leq n^2/4$, $C_2(a,b) \geq n^2/4+n$. Also $d(a,b) \leq (n+2)^3/8$, and shell n contains n+1 weights. Its total contribution is therefore bounded above by $b_n=(n+2)^7 \exp(-tn^2/4)/64$. The Gaussian factor makes the sum of these bounds finite for every $t>0$.

For a triangular cutoff $a+b \leq K$, successive ratios of the shell bound decrease. If the first omitted ratio r is below one, the omitted tail obeys

$$Tail(K) \leq \frac{b_{K+1}}{1-r}, \quad r = \frac{b_{K+2}}{b_{K+1}} < 1.$$

The validation scripts use this bound rather than numerical stabilization alone.

A gauge projector commutes with H_E, so the projected trace is nonnegative and no greater than Z_{full}, but it generally does not factorize. For a fully constrained finite source-free tree it equals 1. At $\beta=0$ the full trace diverges. At fixed positive temperature, $K(t)>1$ and the unprojected product $K(t)^M$ diverges as the link count tends to infinity; finite-volume finiteness is not a finite infinite-volume partition function.

Result 10 Entropy bound for a specified tensor network bond

Let a normalized pure state admit a tree tensor network representation whose cut bond has dimension χ. Across that cut, the reduced-state entropy obeys

$$S(\rho_A) \le \log \operatorname{rank}(\rho_A) \le \log \chi.$$

Proof. Cutting the bond expresses the state as a sum of at most χ product vectors, so its Schmidt rank r is at most χ. The Shannon entropy of its r nonzero Schmidt probabilities is at most log r, with equality for a uniform distribution. If a construction establishes $\chi \le d_\Lambda$, the original bound $S \le \log d_\Lambda$ follows. Equality rank$=d_\Lambda$ is not required.

The existence of a tree-shaped physical interaction graph alone does not impose this bond-rank bound on arbitrary states. Nor does the same reduced-state entropy statement apply to arbitrary mixed states. In a gauge theory, the subsystem algebra or extended-Hilbert-space prescription must also be specified before interpreting this entropy as a physical spatial entanglement measure.

VII Revised quantum subroutines

Subroutine 1 Conditional Gauss projection

On a chosen finite encoding define the positive constraint operator $Q_v=\Sigma_a(G_v^a)^2$. Implement phase estimation on $\exp(i\varphi Q_v)$, using enough ancillas and an evolution scale φ to separate zero from every relevant nonzero eigenvalue modulo 2π. Conditioning on the zero-phase outcome approximates projection onto the singlet subspace.

For exact spectral resolution, the success probability is $\|P_v\psi\|^2$; a state with no invariant component cannot be projected successfully. Finite precision admits leakage, and unsuitable φ can alias a charged state to zero phase. For example, the $3\otimes\bar{3}$ constraint spectrum is $\{0,3\}$; $\varphi=2\pi/3$ makes both eigenphases zero. Postselection, repetition or a specified amplification method is therefore part of a complete procedure. The constraint exponential is not generally a local group transformation $U(g_v)$.

Subroutine 2 A unitary realization of the line test

A traced Wilson operator is not generally unitary and cannot directly serve as a controlled gate. Instead, use the unitary matrix holonomy V_γ acting on a color register, together with a three-dimensional reference prepared in the maximally entangled state Φ. Then

$$\langle\Phi|(V_\gamma \otimes I)|\Phi\rangle = \frac{\operatorname{Tr} V_\gamma}{3}, \quad p_0 = \frac{1}{2}[1 + \operatorname{Re}(\operatorname{Tr} V_\gamma/3)].$$

A Hadamard test on this unitary gives the displayed probability; the factorized Haar vacuum gives average $p_0=1/2$ for a nonempty open path. A quantum implementation must provide the controlled holonomy within its finite link encoding, or a specified normalized block encoding of the desired observable. Sampling or amplitude-estimation error remains part of the measurement cost.

Subroutine 3 Polynomial filtering with a specified encoding

First choose a finite irrep truncation or another bounded operator construction. The full electric Hamiltonian on $L^2(SU(3))$ has infinite operator norm even on one link, so division by its norm is not available without this step. For a bounded Hermitian H, choose a known $\alpha \ge \|H\|$ and implement a unitary block encoding of $A=H/\alpha$.

A QSVT implementation then requires a bounded target polynomial, the appropriate parity, a valid phase sequence and a stated signal convention [3]. A degree-d transformation uses O(d) signal-oracle calls; this does not include the cost of constructing those oracles. Polynomial synthesis, rotation precision and block-encoding error contribute to the total error budget.

For illustration, the bounded filter $F_k(x)=(1-x^2)^k$ preserves x=0 and is at most $(1-\delta^2)^k$ on $\delta\leq|x|\leq1$. To reach tolerance ε, it is enough to take

$$k \geq \frac{log(1/\varepsilon)}{-log(1-\delta^2)}, \quad degree(F_k) = 2k.$$

This explicit filter has $O(\delta^{-2}\log(1/\varepsilon))$ degree and is not claimed to be optimal. It demonstrates why a normalized spectral separation must enter a filtering claim. The paper's former $O(\log(1/\varepsilon))$ statement is meaningful only with fixed spectral promises and an appropriate polynomial construction. Neither the vertex-index width nor a schematic phase product supplies a full application circuit.

VIII Computational validation and significance

The companion package ds_naqha_validation.zip contains exact SU(3) class-integral calculations, rational Casimir calculations, finite matrix tests, Haar Monte Carlo checks, radial Green-function recurrences, partition-tail bounds, entropy calculations and small statevector simulations. The default run uses 20,000 Haar samples with seed 20260929. Its saved report records the environment and tolerances.

Reported category	Count	Interpretation
Passed checks	98	The particular exact or numerical test reproduced its stated target.
Counterexamples	9	A claimed formula or unqualified extension failed in an explicitly constructed case.
Qualifications	21	A necessary assumption, scope restriction or missing specification was recorded.
Unexpected check failures	0	No validation check failed to reproduce its expected outcome.

These 128 report entries are not 128 theorems or independent proofs. Several address different aspects of one result. Counterexamples are expected findings, not software failures. Seven-standard-error Monte Carlo thresholds are diagnostic; they are not certified simultaneous confidence bounds. Floating-point checks and finite models do not prove unrestricted infinite-dimensional statements.

The revised contribution is a precise finite-tree benchmark and an architecture proposal with identified requirements. Gauge projectors, exact electric energies, product-measure invariance and finite-volume trace bounds can support implementation tests. Scalar p-adic boundary scaling is useful context [1,2]. The extension to nontrivial cycles, physically relevant gauge dynamics and practical hardware remains research work.

IX Limitations and research priorities

A meaningful next model should specify its finite graph or quotient core, independent orientations, magnetic cycles, charge sector and boundary constraints before discussing a physical gap. It should then define an increasing-volume limit and establish which spectral quantities remain controlled. Compactness of the projective boundary cannot replace those steps.

A quantum implementation should publish its irrep set, truncation-error analysis, gauge-invariant encoding, oracle circuits, phase synthesis and complete register counts. It should evaluate leakage and approximation errors alongside any reduction in sampling cost. Deterministic polynomial design does not make an experiment error free.

A hardware study should provide a pumped transducer Hamiltonian and measured or explicitly assumed rates, noise and efficiencies. It should compare the heterogeneous design with a single-platform baseline using the same target problem and error tolerance. No claim of demonstrated advantage, unique priority, automatic fault tolerance or four-dimensional Yang–Mills confinement is made.

X References

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[5] DS NAQHA Python validation suite. Version 1.0, 29 September 2026. Files: run_all.py, common.py, validate_foundations.py, validate_theorems.py, validate_subroutines.py, and saved validation reports. Companion artifact: ds_naqha_validation.zip.

Appendix Revision record

This edition updates the September 2026 source paper and preserves the numbering of its ten mathematical results. It replaces the blanket mass-gap and infinite-volume discreteness language with sector-specific finite-volume statements; corrects the cycle Haar-integral argument; supplies the Green-function sign and mass conventions; adds the degenerate first-order counterexample; separates binary indexing from one-hot and oracle costs; imposes positive temperature and distinguishes projected traces; and adds the tensor-network bond hypothesis to the entropy bound.

The algorithmic section now states QPE aliasing and success conditions, replaces a nonunitary controlled trace by a valid unitary realization, and makes QSVT truncation, polynomial and gap requirements explicit. The hardware section retains the proposed platform allocation while requiring a quantitative pumped-conversion model. The bibliography lists references supporting the revised exposition rather than carrying forward unexamined priority claims.

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